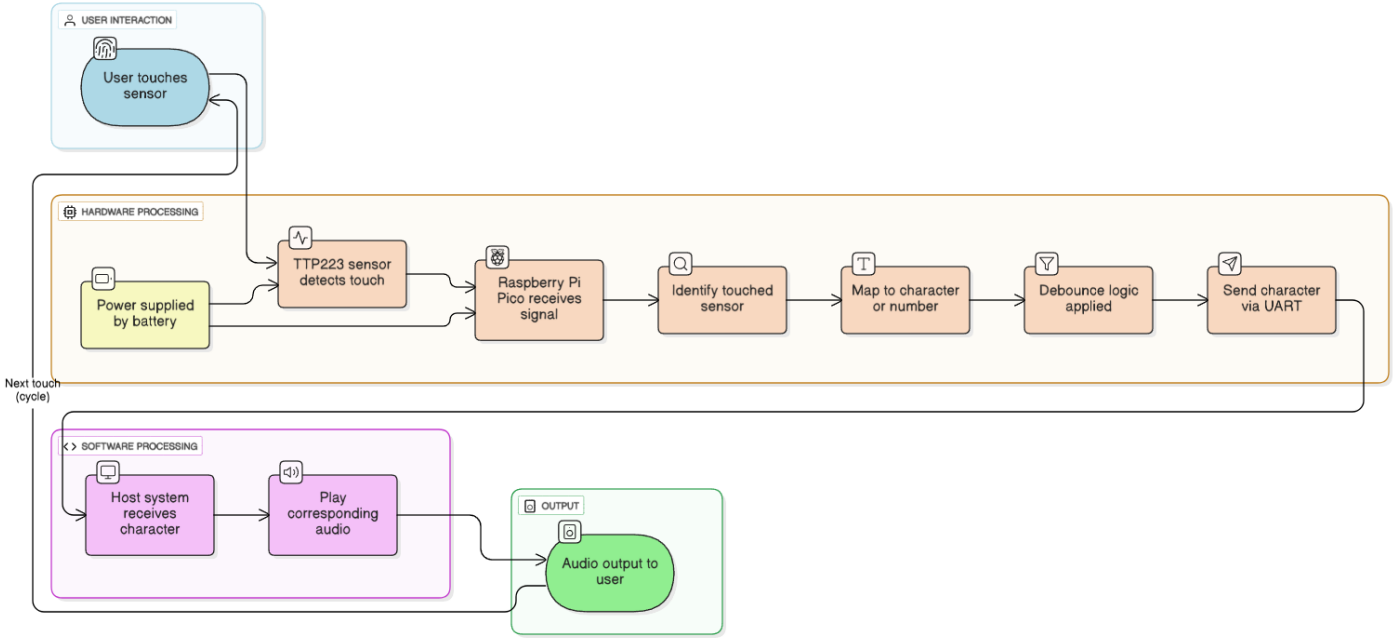
System Architecture



1) User-Level Overview

The *BRAILLEAR* system enables visually impaired users to learn alphabets and numbers through **touch and sound interaction**.
When a user touches a sensor, the system detects it, processes it, and plays the corresponding sound instantly — creating a **tactile-to-audio learning experience**.

2) Hardware Flow

- **Touch Input (TTP223 Sensors)**
  - The user touches one of the 26 TTP223 capacitive touch sensors, each representing a letter (A–Z).

- **Signal Processing (Raspberry Pi Pico)**
  - The microcontroller detects which sensor is touched through its GPIO pins.
  - It then maps the signal to a specific alphabet or number.

- **Data Transmission (UART Communication)**
  - The Raspberry Pi Pico sends the detected character to the laptop via UART serial connection.

- **Power Supply**

- A rechargeable 3.7V–5V battery powers the entire system, ensuring portability.

3) Software Flow

- **Microcontroller Code (MicroPython)**
  - Detects the sensor input and applies debounce logic to remove noise.
  - Sends the recognized character through the serial port.

- **Host System Code (Python Application)**
  - Receives the character data from the microcontroller using the *PySerial* library.
  - Uses the *PyDub* library to play the audio file corresponding to that character (A–Z or 0–9).

- **Audio Output**
  - The speaker or headphone outputs the correct pronunciation of the touched letter, completing the learning cycle.